

9.3 Resistivity

Question Paper

Course	CIE A Level Physics
Section	9. Electricity
Topic	9.3 Resistivity
Difficulty	Hard

Time allowed: 20

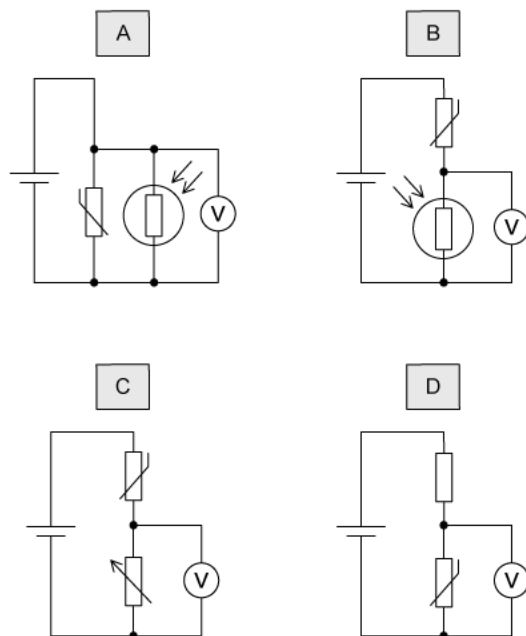
Score: /10

Percentage: /100

Question 1

A thermistor and another component are connected to a constant voltage supply. A voltmeter is connected across one of the components. The temperature of the thermistor is then reduced, but no other changes are made.

In which circuit will the voltmeter reading increase?



[1 mark]

Question 2

A copper wire is to be replaced by an aluminum alloy wire of the same length and resistance.

Copper has half the resistivity of the alloy.

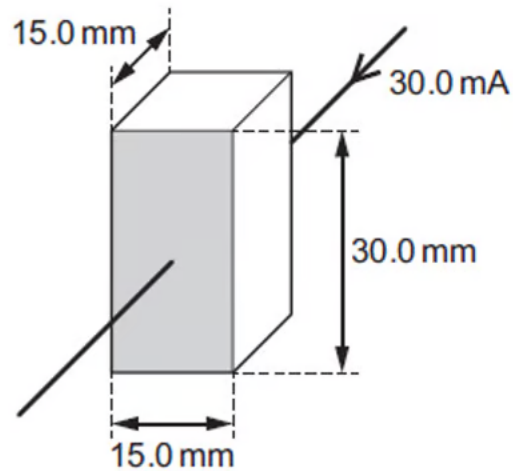
What is the ratio $\frac{\text{diameter of alloy wire}}{\text{diameter of copper wire}}$?

- A. $\sqrt{2}$
- B. 2
- C. $\sqrt[2]{2}$
- D. 4

[1 mark]

Question 3

The current in a block of semiconductor is 30.0 mA when there is a potential difference (p.d.) of 10.0 V across it. The dimensions of the block and the direction of the current in it are as shown.



The electrical meters used are accurate to ± 0.1 mA and ± 0.1 V. The dimensions of the block are accurate to ± 0.2 mm.

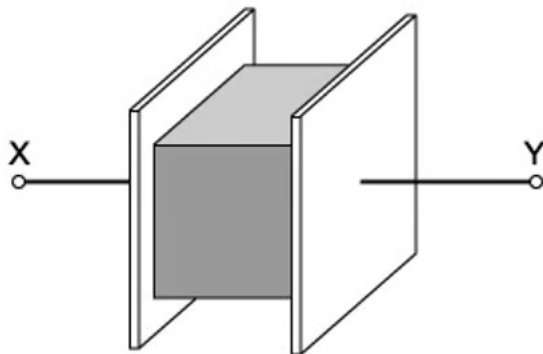
What is the resistivity of the semiconductor?

- A. $10.0 \pm 0.2 \Omega \text{ m}$
- B. $10.0 \pm 0.3 \Omega \text{ m}$
- C. $10.0 \pm 0.5 \Omega \text{ m}$
- D. $10.0 \pm 0.8 \Omega \text{ m}$

[1 mark]

Question 4

The resistance of a metal cube is measured by placing it between two parallel plates, as shown.



The cube has volume V and is made of a material with resistivity ρ . The connections to the cube have negligible resistance.

Which expression gives the electrical resistance of the metal cube between X and Y?

A. $\rho V^{\frac{1}{3}}$

B. $\rho V^{\frac{2}{3}}$

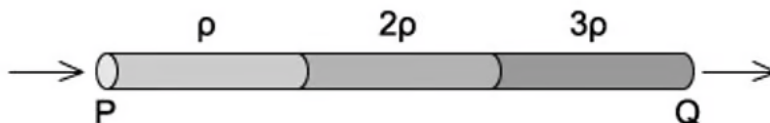
C. $\frac{\rho}{V^{\frac{1}{3}}}$

D. $\frac{\rho}{V^{\frac{2}{3}}}$

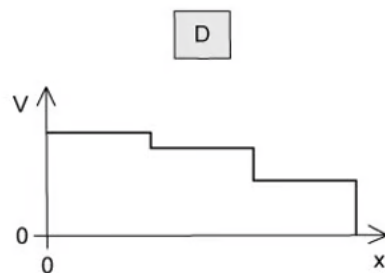
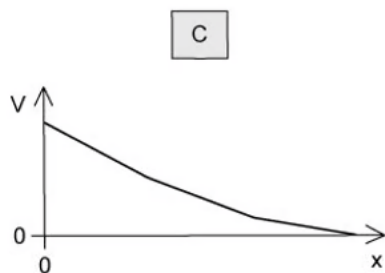
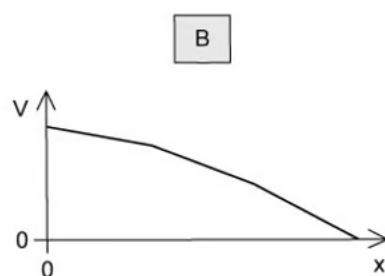
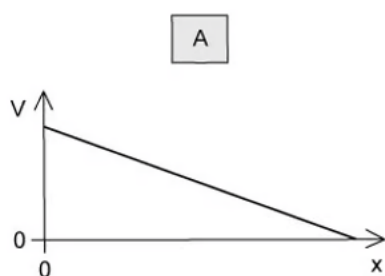
[1 mark]

Question 5

A wire PQ is made of three different materials, with resistivities ρ , 2ρ and 3ρ . There is a current I in this composite wire, as shown.



Which graph best shows how the potential V along the wire varies with distance x from P?



[1 mark]

Question 6

A strain gauge consists of a length of wire with a uniform cross-sectional area. Its resistance is $2.50\text{ k}\Omega$. It is attached to a large pipe.

When the pipe expands, the strain gauge changes in dimensions, such that the length increases by 4% and the diameter reduced by 1%.

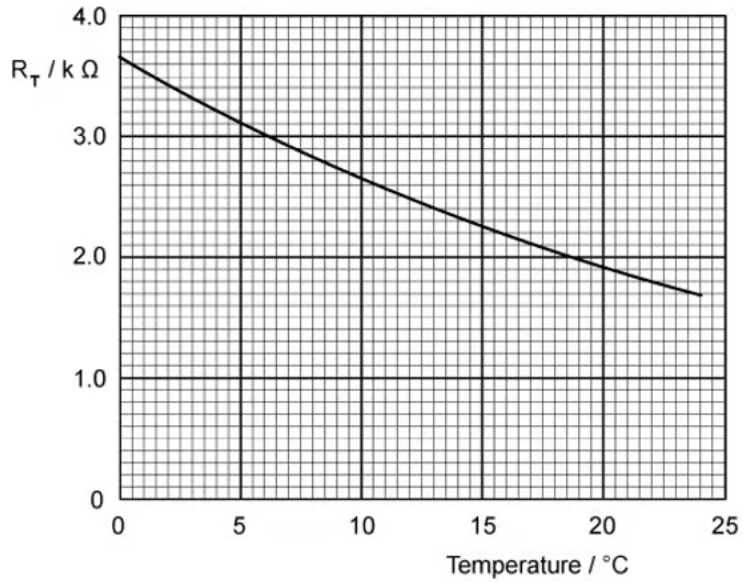
What is the new resistance of the strain gauge?

- A. $2.60\text{ k}\Omega$
- B. $2.65\text{ k}\Omega$
- C. $2.80\text{ k}\Omega$
- D. $3.00\text{ k}\Omega$

[1 mark]

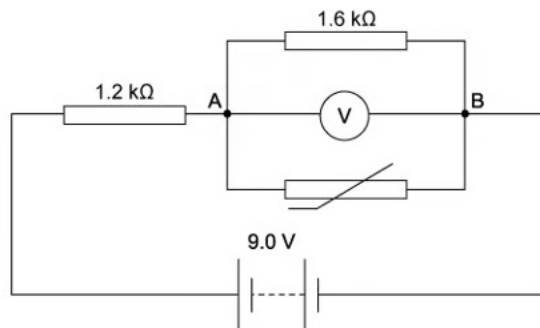
Question 7

The variation with temperature of the resistance R_T of a thermistor is shown on the graph.



The thermistor is connected into the circuit shown below.

The battery has an e. m. f. of 9.0 V and negligible internal resistance. The voltmeter has infinite resistance.



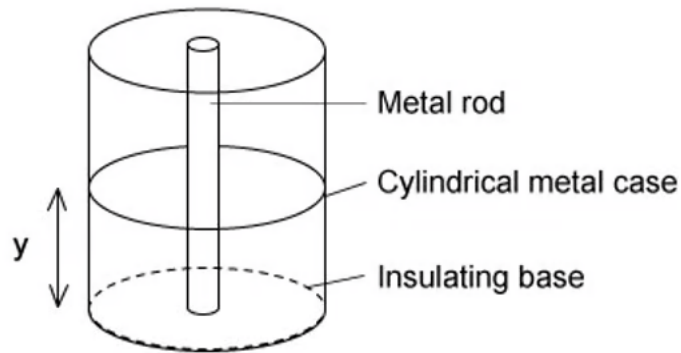
What is the temperature of the thermistor when the reading on the voltmeter is 4.0 V?

- A. 2.5 °C
- B. 13.0 °C
- C. 16.0 °C
- D. 24.0 °C

[1 mark]

Question 8

A conducting liquid fills a cylinder metal case to a depth y as shown in the diagram.



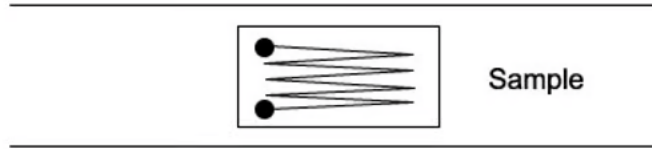
The resistance between the case and the metal rod is

- A. proportional to y
- B. proportional to y^2
- C. inversely proportional to y
- D. inversely proportional to y^2

[1 mark]

Question 9

The following strain gauge consists of a fine metal wire, so that an applied tension on the strain gauge results in changes in the length and cross-sectional area of the wire.



When the length of the wire is stretched by 5%, what is the percentage change in the resistance of the wire? Assume that the volume of a wire remains constant when it is stretched.

- A. + 5%
- B. – 5%
- C. + 10%
- D. – 10%

[1 mark]**Question 10**

Aluminum and copper rods are designed to have the same length and the same resistance.

The resistivity of copper is half that of aluminum, but its density is three times that of aluminum.

Determine the ratio of the mass of the aluminum rod to the mass of the copper rod.

- A. $\frac{1}{6}$
- B. $\frac{2}{3}$
- C. $\frac{3}{2}$
- D. $\frac{6}{1}$

[1 mark]

Model Answers: Hard

1

The correct answer is **D** because:

- The resistance of a thermistor decreases as the temperature increases
- The temperature of the thermistor is reduced, and no other changes are made, so, its resistance increases
- From Ohm's law, $V = IR$, an increase in the resistance of the thermistor would cause the p.d. across it to increase
- Therefore, for the voltmeter reading to increase, the voltmeter should be connected in parallel to the thermistor so that it reads the p. d. across the thermistor

A is incorrect because the voltmeter reading remains constant since all three components are connected in parallel to each other

B is incorrect because the resistance of an LDR decreases as the light intensity increases, but in this case, the light intensity is unchanged, so its resistance is unchanged

C is incorrect because the voltmeter is across the variable resistor so the increase in resistance from the thermistor would cause a decrease in resistance across the variable resistor

2

The correct answer is **A** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area of a wire is equal to $A = \pi r^2 = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$
 - The resistance of the copper wire is equal to $R = \frac{\rho_c L}{A}$

- So, the cross-sectional area is equal to $A = \frac{\pi d_c^2}{4}$
- We can combine the above equations to give
 - $R = \rho_c L \left(\frac{4}{\pi d_c^2} \right)$
- So, the diameter of the copper wire is equal to
 - $d_c^2 = \frac{4\rho_c L}{\pi R}$
 - $d_c = 2\sqrt{\frac{\rho_c L}{\pi R}}$
- We are told copper has half the resistivity of the alloy, so the resistivity of the alloy is equal to $2\rho_c$
 - The resistance of the aluminum wire is equal to $R = \frac{(2\rho_c)L}{A}$
 - So, the cross-sectional area is equal to $A = \frac{\pi d_A^2}{4}$
- We can combine these equations to give
 - $R = 2\rho_c L \left(\frac{4}{\pi d_A^2} \right)$
- So, the diameter of the aluminum wire is equal to
 - $d_A^2 = \frac{8\rho_c L}{\pi R}$
 - $d_A = 2\sqrt{2}\sqrt{\frac{\rho_c L}{\pi R}}$
- So, the ratio $\frac{d_A}{d_c} = \frac{2\sqrt{2}}{2} = \sqrt{2}$

3

The correct answer is **C** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- So, resistivity is equal to $\rho = \frac{AR}{L}$
- From Ohm's law, resistance is equal to $R = \frac{V}{I}$
- So, combining this with the equation for resistivity, we get

- $\rho = \frac{VA}{IL}$
- The cross-sectional area of the block is equal to the area of the grey rectangle through which the current flows
 - $A = lw$, where l is the length and w is the width
- So, resistivity is equal to $\rho = \frac{Vlw}{IL}$
 - The p. d. is equal to $V = 10.0 \text{ V}$
 - The current through the block is equal to $I = 30 \text{ mA} = 30 \times 10^{-3} \text{ A}$
- The dimensions of the block are equal to
 - $l = 15 \times 10^{-3} \text{ m}$, $w = 30 \times 10^{-3} \text{ m}$, $L = 15 \times 10^{-3} \text{ m}$
- So, resistivity $\rho = \frac{Vlw}{IL} = \frac{10 \times (15 \times 10^{-3}) \times (30 \times 10^{-3})}{(30 \times 10^{-3}) \times (15 \times 10^{-3})} = 10 \Omega \text{m}$
- Next, we must consider the uncertainty in resistivity ρ
 - $\frac{\Delta \rho}{\rho} = \left(\frac{\Delta V}{V} \right) + \left(\frac{\Delta l}{l} \right) + \left(\frac{\Delta w}{w} \right) + \left(\frac{\Delta I}{I} \right) + \left(\frac{\Delta L}{L} \right)$
 - $\Delta \rho = \left[\left(\frac{\Delta V}{V} \right) + \left(\frac{\Delta l}{l} \right) + \left(\frac{\Delta w}{w} \right) + \left(\frac{\Delta I}{I} \right) + \left(\frac{\Delta L}{L} \right) \right] \times \rho$
 - $\Delta \rho = \left[\left(\frac{0.1}{10} \right) + \left(\frac{0.2}{15} \right) + \left(\frac{0.2}{30} \right) + \left(\frac{0.1}{30} \right) + \left(\frac{0.2}{15} \right) \right] \times 10$
 - $\Delta \rho = 0.467 = 0.5 \Omega \text{m}$
- Therefore, resistivity is equal to $\rho = 10.0 \pm 0.5 \Omega \text{m}$

4

The correct answer is **C** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Where L is the length of the wire and A is the cross-sectional area of the wire
- For a cube, all the sides are of equal lengths
- Let the length of a side be L
 - So, the volume of the cube is $V = L^3$
 - The cross-sectional area in contact with the plates is equal to $A = L^2$
- Resistance is therefore equal to $R = \frac{\rho L}{L^2} = \frac{\rho}{L}$
- But since $V = L^3$, length L is equal to $L = \sqrt[3]{V} = V^{\frac{1}{3}}$
 - So, resistance $R = \frac{\rho}{V^{\frac{1}{3}}}$

Exam Tip: An alternative (and potentially quicker) method is to consider the units of the quantities involved, only choice **C** gives the unit of resistance (Ω):

- Unit of resistivity = Ωm
- Unit of volume $V = \text{m}^3$
- Unit of $\frac{\rho}{V^{\frac{1}{3}}} = [\Omega\text{m}][\text{m}^3]^{-\frac{1}{3}} = [\Omega\text{m}]\text{m}^{-1} = \Omega$

5

The correct answer is **B** because:

- Ohm's law tells us potential difference is equal to $V = IR$
- The resistance R of a wire is proportional to the resistivity ρ of the material from which the wire is made $R = \frac{\rho L}{A}$
- When the resistivity of the material is greater, its resistance is also greater and therefore, the potential difference across it is greater
 - So, the potential difference is greatest across the 3ρ material
 - On the graph, this will be represented by three separate lines with increasing steepness
- Note that the graph gives the potential V along the wire on the y-axis while the V from represents the potential difference – which is the change in potential
- The change in potential ΔV as the distance changes Δx is represented by the gradient, which is constant for each value of ρ
- Resistance of the wire also depends on the length of wire being considered. So, as x increases, the length of wire being considered increases and so, as x increases, the difference in potential V at that particular value of x is greater
 - This means in general, as x increases, V decreases, even if the resistivity is constant
- Therefore, the only graph matching this description is option B

A is incorrect because the gradient is constant everywhere

C is incorrect because the gradient of the straight lines decreases as x increases

D is incorrect because the gradient is zero

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D is incorrect because the gradient is zero

6

The correct answer is **B** because:

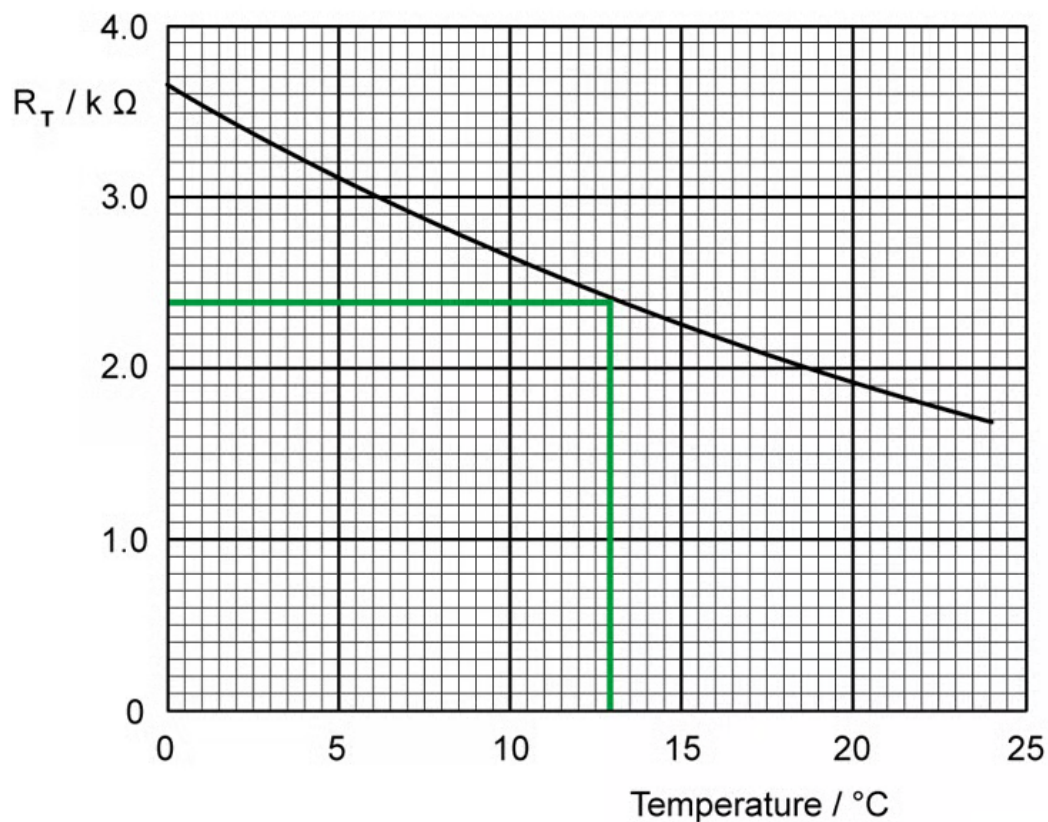
- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area of a wire is equal to $A = \pi r^2 = \pi \left(\frac{d}{2}\right)^2 = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$
- When the pipe expands
 - The new length is given by $L' = (1 + 0.04)L = 1.04L$
 - The new cross-sectional area is given by $A' = (1 - 0.01)^2 A = (0.99)^2 A$ (squared because A is proportional to d^2)
 - The resistivity ρ doesn't change since the material stays the same
- So, the new resistance is equal to
 - $R' = \frac{\rho L'}{A'} = \frac{1.04\rho L}{(0.99)^2 A} = \frac{1.04}{(0.99)^2} R = \frac{1.04}{(0.99)^2} \times 2.5 = 2.653 \text{ k}\Omega$
- Therefore, the new resistance is 2.65 k Ω

7

The correct answer is **B** because:

the correct answer is 11.0 °C.

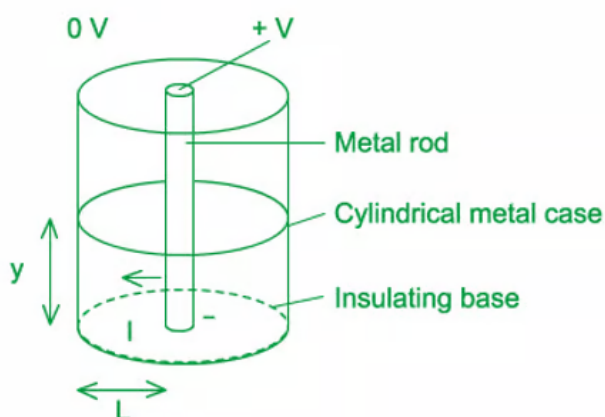
- Ohm's law states $V = IR$, V is directly proportional to resistance R
- We are told the reading on the voltmeter is $= 4.0 \text{ V}$
- So, the p.d. across the $1.2 \text{ k}\Omega$ is $9 - 4 = 5.0 \text{ V}$
- A p.d. of 5.0 V corresponds to $1.2 \text{ k}\Omega$
 - So, 4.0 V would correspond to a resistance of $\frac{4.0 \text{ V}}{5.0 \text{ V}} \times 1200 \Omega$
- The total resistance between A and B $= \frac{4}{5} \times 1200 = 960 \Omega$
- The thermistor and $1.6 \text{ k}\Omega$ resistor are in parallel so
 - $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
 - $\frac{1}{960} = \frac{1}{1600} + \frac{1}{R_T}$
 - $\frac{1}{R_T} = \frac{1}{2400}$
- Therefore, the resistance of the thermistor is $2.4 \text{ k}\Omega$
- From the graph, the temperature that corresponds to $2.4 \text{ k}\Omega$ is $11.0 \text{ }^\circ\text{C}$



8

The correct answer is **C** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- When a potential difference is applied across the metal rod and the case, current /will flow through the liquid as shown
- The current will flow over a length L and through a cross-sectional area A
- In this case, the cross-sectional area is the surface area of the cylinder ($2\pi rh$), where h is equal to the depth y
- So, the cross-sectional area A is proportional to y
 - Therefore, R is inversely proportional to y



9

The correct answer is **C** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Let's call the new length and cross-sectional area L' and A' respectively
- Since the volume remains constant
 - $V = AL = A'L'$
- When the wire expands
 - The new length is given by $L' = (1 + 0.05)L = 1.05L$
 - The new cross-sectional area is given by $V = AL = (1.05L)\left(\frac{A}{1.05}\right)$
 - So, $A' = \frac{A}{1.05}$

1.05

- The resistivity ρ doesn't change since the material stays the same
- So, the new resistance is equal to
 - $R' = \frac{\rho L'}{A'} = \rho(1.05) \left(\frac{1.05}{A} \right) = (1.05)^2 \left(\frac{\rho L}{A} \right) = 1.10R$
- Therefore, the percentage change in R is equal to
 - $(1.10 - 1.00) \times 100\% = 10\%$

10

The correct answer is **B** because:

- In this question we will refer to density by d and resistivity by ρ
- Density of a material is given by $d = \frac{m}{V}$
 - The mass of the aluminium rod is $m_A = d_A V_A$
 - The mass of the copper rod is $m_c = d_c V_c$
- The volume of a rod is given by $V = AL$
 - The volume of the aluminium rod is $V_A = A_A L$
 - The volume of the copper rod is $V_c = A_c L$
- The ratio of mass of aluminium to mass of copper is equal to
 - $\frac{m_A}{m_c} = \frac{d_A V_A}{d_c V_c} = \frac{d_A A_A L}{d_c A_c L} = \frac{d_A A_A}{d_c A_c}$
- Since $d_c = 3d_A$ (the density of copper is three times that of aluminium)
 - $\frac{m_A}{m_c} = \frac{d_A A_A}{3d_A A_c} = \frac{1}{3} \left(\frac{A_A}{A_c} \right)$
- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Since the resistance and length of the rods are the same, $\frac{\rho}{A}$ is constant, so:
 - $\frac{\rho_c}{A_c} = \frac{\rho_A}{A_A}$
 - $\frac{A_A}{A_c} = \frac{\rho_A}{\rho_c}$
- Since $\rho_A = 2\rho_c$ (the resistivity of copper is half that of aluminium)

$$\circ \quad \frac{A_A}{A_c} = \frac{\rho_A}{\rho_C} = \frac{2}{1}$$

- Therefore, the ratio $\frac{\text{mass of aluminium}}{\text{mass of copper}} = \frac{m_A}{m_c} = \frac{1}{3} \left(\frac{A_A}{A_c} \right) = \frac{1}{3} \times \frac{2}{1} = \frac{2}{3}$

